Final Project Proposal Overview

Advanced Embedded Software Development

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Learning objectives:

How to propose your final project.

How to retarget your project for hardware instead of QEMU.

Final Project Proposal Overview

- See details in Coursera Reading Assignment
- Will include peer review portion - each student will peer review 3 other student projects
  - Your peer review is assigned randomly
  - You are graded on completing your peer reviews
- Suggestions for Sprint 1 Target:
  - Buildroot/yocto build complete for hardware platform.
  - Some software work completed.
  - Buildroot/yocto setup steps completed, board booting and communicating successfully.

Final Project Preview

- Implemented on your own or organized in groups of up to 3.
- Requirements:
  - Uses real hardware
  - Use Buildroot or Yocto
  - Has some relationship to the content we covered in the course sequence
    - If you don't have ideas about what to do, think about what we've covered in class which hasn't been covered in assignments.
    - Also look at previous projects and "future work" suggestions in projects linked from https://sites.google.com/colorado.edu/ecen5713/home

Project Prep: Retargeting for Hardware

- See details in hardware platform linked from https://github.com/cu-ecen-aeld/aesd-assignments/wiki/Supported-Hardware-Platforms
  - Modify Buildroot or Yocto configuration to support building on hardware and/or QEMU
  - Use clean builds when switching projects
    - Alternatively, use different directories for build targeting qemu or devices.
  - Flash the image to an SDCard
    - Use dd on Linux
    - Use Rufus or similar on Windows

Final Project Schedule Overview

- Consists of 3 checkpoints, 1 which we will refer to as “Sprints” using Agile terminology
- Each should contain the rough equivalent of one assignment worth of effort per project person (~10-15 hours of work)
- You will plan the work for the next sprint in the previous sprint
- You will use github issues to track and report your work
- You will use github wiki pages to report overall progress/status

Final Project Schedule

Item
Final Project Proposal
Sprint 1
Sprint 2
Sprint 3
Final Project Demo Outline and Video

Final Project Scrum

- Session held with your teammates for multi-person projects
- What did you do since last report?
- What are you going to do before the next report?
- Any blockers?

Final Project Idea Suggestions

- Consider something we haven't covered fully in class/assignments.
- Look for projects which have done similar things online.
  - Possibly porting to Yocto/Buildroot
  - Possibly using a different hardware platform
- Have a plan for implementation of major project components
  - Don't assume you will figure it out later.
- Look at possibly useful project components at https://github.com/cu-ecen-aeld/aesd-assignments/wiki/Supported-Hardware-Platforms
- Look at past projects https://sites.google.com/colorado.edu/ecen5713/home#h.p_nEXFZJYmVRsc

Buildroot:Retargeting for Hardware

- See https://github.com/cu-ecen-5013/buildroot-assignments-base/wiki/Setting-up-Buildroot-for-Hardware-Builds
- Modify shared.sh to use hardware configuration
- ./clean.sh
- ./build.sh
  - Will use AESD_DEFAULT_DEFCONFIG
- ./save_config.sh
  - Save .config based on AESD_DEFAULT_DEFCONFIG to AESD_MODIFIED_DEFCONFIG

Buildroot: Retargeting for Hardware

- buildroot menuconfig customization to add packages back into this build
  - dropbear for SSH
  - Set root password
  - Add assignment/ldd packages
  - Setup rootfs overlay
- ./save_config.sh
  - Save updates to AESD_MODIFIED_DEFCONFIG
- ./build.sh - start the new build for this platform

Hardware Debug/Deployment Strategies

- Re-flashing the image is time consuming.
- Alternatively:
  - Test in QEMU

    or
  - Copy executables from buildroot output/build directory to target filesystem using scp for test purposes.
- Rebuild/re-test complete image when successful